

Association Rule Learning on the Groceries Dataset

Assignment: Association Rule Learning

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Code repository:

https://github.com/charerimana/data-mining-labs/groceries_association_rules.ipynb

Software: Python 3 (pandas, mlxtend, matplotlib) / Algorithm: Apriori

Overview

This report applies **association rule learning** to the Groceries dataset to discover which products customers tend to buy together. The data is in transaction (basket) format: each line is one shopping trip listing the items purchased. The analysis was carried out in Python using the **Apriori** algorithm from the **mlxtend** library. Three measures are used throughout: **support** (how often an itemset appears), **confidence** (how reliable a rule is), and **lift** (how much stronger the association is than chance).

Question 1 - Understand the Data

a) **Number of transactions:** 9,835

b) **Number of distinct grocery items:** 169

c) **The 10 most frequently purchased items** are shown in Table 1. *Count* is the number of baskets containing the item; *support* is that count divided by 9,835.

Table 1: Top 10 most frequently purchased items.

Rank	Item	Count	Support
1	whole milk	2,513	0.2555
2	other vegetables	1,903	0.1935
3	rolls/buns	1,809	0.1839
4	soda	1,715	0.1744
5	yogurt	1,372	0.1395
6	bottled water	1,087	0.1105
7	root vegetables	1,072	0.1090
8	tropical fruit	1,032	0.1049
9	shopping bags	969	0.0985
10	sausage	924	0.0940

Whole milk is by far the most common item, appearing in about 1 in 4 baskets, followed by other vegetables and rolls/buns. Even the top item has a support of only about 0.26, which shows that no single product is bought by most customers - grocery baskets are very varied.

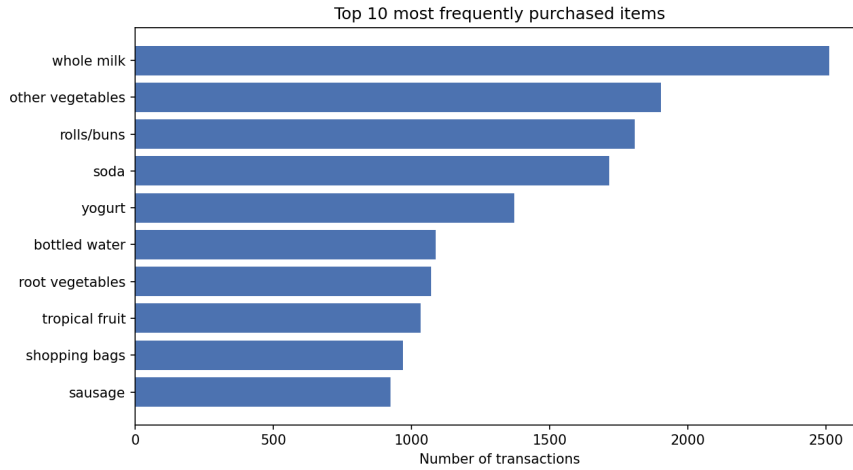


Figure 1: Top 10 most frequently purchased items.

Question 2 - Frequent Itemsets

Using Apriori with a minimum support of 0.01 (an itemset must appear in at least 1% of baskets, i.e. about 98 trips), 245 frequent itemsets of two or more items were found. The 10 with the highest support are listed in Table 2 (b - support values included).

Table 2: Top 10 frequent itemsets containing two or more items.

#	Itemset (2 items)	Support
1	{other vegetables, whole milk}	0.0748
2	{rolls/buns, whole milk}	0.0566
3	{whole milk, yogurt}	0.0560
4	{root vegetables, whole milk}	0.0489
5	{other vegetables, root vegetables}	0.0474
6	{other vegetables, yogurt}	0.0434
7	{other vegetables, rolls/buns}	0.0426
8	{tropical fruit, whole milk}	0.0423
9	{soda, whole milk}	0.0401
10	{rolls/buns, soda}	0.0383

c) Highest support: {other vegetables, whole milk} with a support of 0.0748 - this pair appears together in about 7.5% of all transactions (roughly 736 baskets).

d) What support tells us: Support measures how popular a combination is across all customers. A support of 0.0748 means that out of every 100 shopping trips, roughly 7-8 include both other vegetables and whole milk. High-support itemsets represent common, everyday buying patterns (staples people buy together), so they are the combinations a store sees most often.

Question 3 - Association Rules

From the frequent itemsets, 170 rules were generated with a minimum confidence of 0.25. Twelve representative rules (sorted by lift) are reported in Table 3, each with its antecedent, consequent, support, confidence and lift.

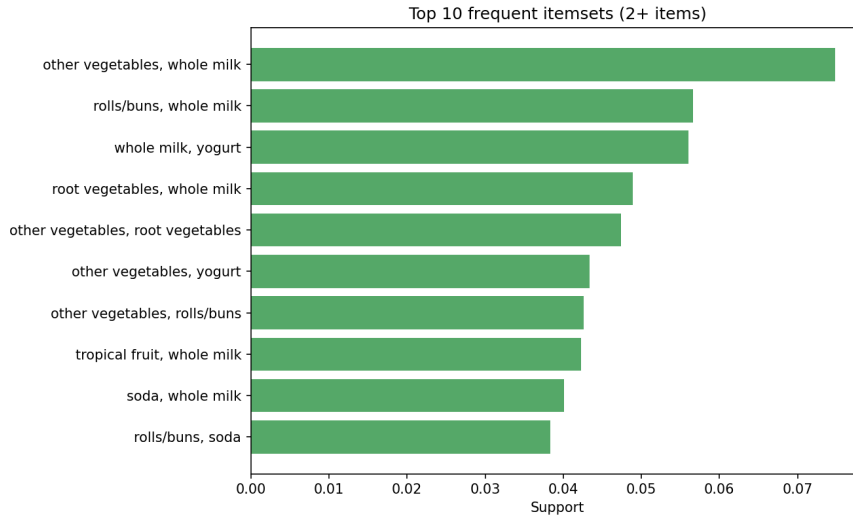


Figure 2: Top 10 frequent itemsets (2+ items) by support.

Table 3: Twelve representative association rules (sorted by lift).

Antecedent	Consequent	Support	Confidence	Lift
{citrus fruit, other vegetables}	{root vegetables}	0.0104	0.359	3.295
{other vegetables, tropical fruit}	{root vegetables}	0.0123	0.343	3.145
{beef}	{root vegetables}	0.0174	0.331	3.040
{citrus fruit, root vegetables}	{other vegetables}	0.0104	0.586	3.030
{root vegetables, tropical fruit}	{other vegetables}	0.0123	0.585	3.021
{other vegetables, whole milk}	{root vegetables}	0.0232	0.310	2.842
{curd, whole milk}	{yogurt}	0.0101	0.385	2.761
{other vegetables, yogurt}	{root vegetables}	0.0129	0.297	2.729
{other vegetables, yogurt}	{tropical fruit}	0.0123	0.283	2.701
{other vegetables, rolls/buns}	{root vegetables}	0.0122	0.286	2.628
{tropical fruit, whole milk}	{root vegetables}	0.0120	0.284	2.602
{rolls/buns, root vegetables}	{other vegetables}	0.0122	0.502	2.595

Question 4 - Understanding the Measures

Three rules are explained below.

Rule A: {other vegetables, whole milk} -> {root vegetables} (supp 0.0232, conf 0.310, lift 2.842)

- **Support (0.0232):** about 2.3% of all baskets contain all three items together - other vegetables, whole milk and root vegetables.
- **Confidence (0.310):** of the customers who bought other vegetables and whole milk, 31% also bought root vegetables.
- **Lift (2.842):** those customers are about 2.8 times more likely to buy root vegetables than a random customer - a strong positive association.

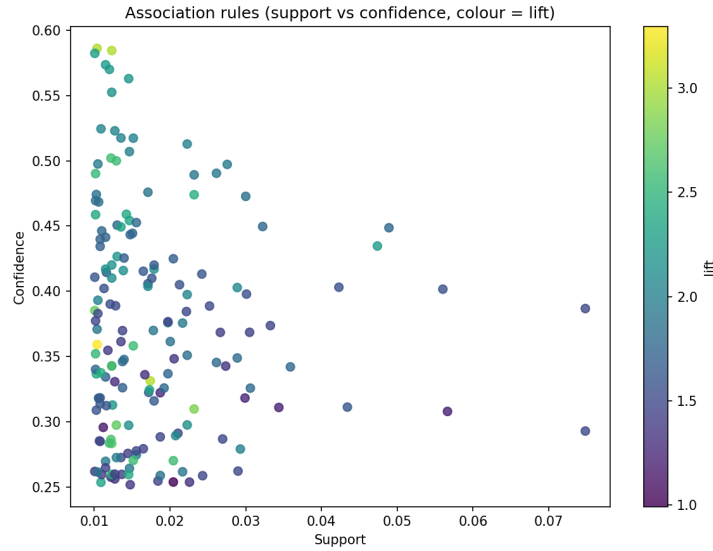


Figure 3: All rules plotted by support and confidence, coloured by lift.

Rule B: {citrus fruit, root vegetables} -> {other vegetables} (supp 0.0104, conf 0.586, lift 3.030)

- **Support (0.0104):** about 1% of all baskets contain citrus fruit, root vegetables and other vegetables together.
- **Confidence (0.586):** when a basket already has citrus fruit and root vegetables, 58.6% of the time it also has other vegetables - a very reliable rule.
- **Lift (3.030):** these customers are about 3 times more likely to buy other vegetables than average.

Rule C: {curd, whole milk} -> {yogurt} (supp 0.0101, conf 0.385, lift 2.761)

- **Support (0.0101):** about 1% of baskets contain curd, whole milk and yogurt together.
- **Confidence (0.385):** 38.5% of customers who buy curd and whole milk also buy yogurt.
- **Lift (2.761):** buying curd and whole milk makes yogurt about 2.8 times more likely than chance - an intuitive dairy-with-dairy pattern.

Question 5 - Find Interesting Rules

Table 4: The most interesting rules by each measure.

Criterion	Rule	Supp.	Conf.	Lift
Highest confidence	{citrus fruit, root vegetables} -> {other vegetables}	0.0104	0.586	3.030
Highest lift	{citrus fruit, other vegetables} -> {root vegetables}	0.0104	0.359	3.295
Highest support	{whole milk} -> {other vegetables}	0.0748	0.293	1.514

- Highest confidence:** {citrus fruit, root vegetables} -> {other vegetables} (0.586).
- Highest lift:** {citrus fruit, other vegetables} -> {root vegetables} (3.295).
- Highest support:** {whole milk} -> {other vegetables} (0.0748).

d) Are they the same rule? No. They are three different rules, and this is expected because the three measures capture different things. Support rewards popular items, so the highest-support rule involves whole milk and other vegetables - the two best-selling products - even though its lift (1.51) is modest. Confidence rewards reliability, and lift rewards surprise/strength relative to chance; these favour the vegetable-and-fruit combinations, which are less common overall (lower support) but very strongly linked when they do occur. A good rule is usually one that scores reasonably on all three, not just one.

Question 6 - Interpret a Rule

Selected rule: {root vegetables, tropical fruit} → {other vegetables} (support 0.0123, confidence 0.585, lift 3.021).

In plain language: customers who buy root vegetables and tropical fruit together also tend to buy other vegetables in the same trip.

Is it strong or weak? This is a **strong** association. Its confidence is 0.585, meaning almost 6 out of 10 customers who buy root vegetables and tropical fruit also add other vegetables to their basket. More importantly, its lift is 3.02 - well above 1 - so this behaviour is about three times more likely than if the purchases were unrelated. The pattern also makes practical sense: these are all fresh-produce items, so a shopper stocking up on fruit and vegetables naturally buys across the whole produce section. The only limitation is its support (0.0123 (about 1.2%) of baskets), so while the link is strong, it applies to a modest slice of all customers.

Question 7 - Business Application (Supermarket Manager)

a) Two products to place close together: other vegetables and whole milk. They form the most frequent pair (support 0.0748), so placing them near each other - or on the natural walking path between them - makes it easy for the many customers who buy both and encourages the pairing.

b) A possible product bundle: a "Fresh Produce" bundle of root vegetables + tropical fruit + other vegetables, based on the strong rule in Question 6 (confidence 0.585, lift 3.02). A small discount on the third item when the first two are bought would suit these shoppers.

c) Using rules to recommend products: at checkout or in an app, when a customer's basket already contains curd and whole milk, the system can recommend yogurt (Rule C, lift 2.76), because such customers are about 2.8 times more likely to want it. The same idea powers "customers who bought this also bought..." suggestions and targeted coupons printed on receipts.

Question 8 - Understanding Association

a) Does a rule prove causation? No. Association rules only show that products are bought together more often than expected; they do not prove that buying one *causes* buying the other. The link may be caused by a third factor (e.g. a customer planning a specific meal, a promotion, or shelf placement), or it may simply be a co-occurrence of two popular items. Association is correlation, not cause.

b) Why can a rule have high confidence but low lift? Confidence can be high simply because the consequent is very popular on its own. For example, many rules point to whole milk with decent confidence only because whole milk is in 25% of all baskets - so a lot of customers buy it regardless of what else is in the cart. If the consequent's overall support is already high, the lift (which divides confidence by that baseline support) stays close to 1, meaning the rule adds little real information even though its confidence looks good.

c) What does lift > 1 indicate? A lift greater than 1 means the two sides of the rule appear together more often than would be expected if they were independent - a positive association. The

higher the lift, the stronger the link. (A lift of exactly 1 means independence, and a lift below 1 means the items are bought together less often than chance, i.e. a negative association.)

Appendix - Method Notes

- Data: 9,835 transactions, 169 unique items, transaction (basket) format, loaded from `data/groceries.csv`.
- Tool: Python 3 with pandas, mlxtend (`TransactionEncoder`, `apriori`, `association_rules`) and matplotlib.
- Frequent itemsets: Apriori, minimum support = 0.01.
- Rules: minimum confidence = 0.25; 170 rules produced. Full rule table available in `q3_association_rules.csv`.
- Code repository: https://github.com/charerimana/data-mining-labs/groceries_association_rules.ipynb